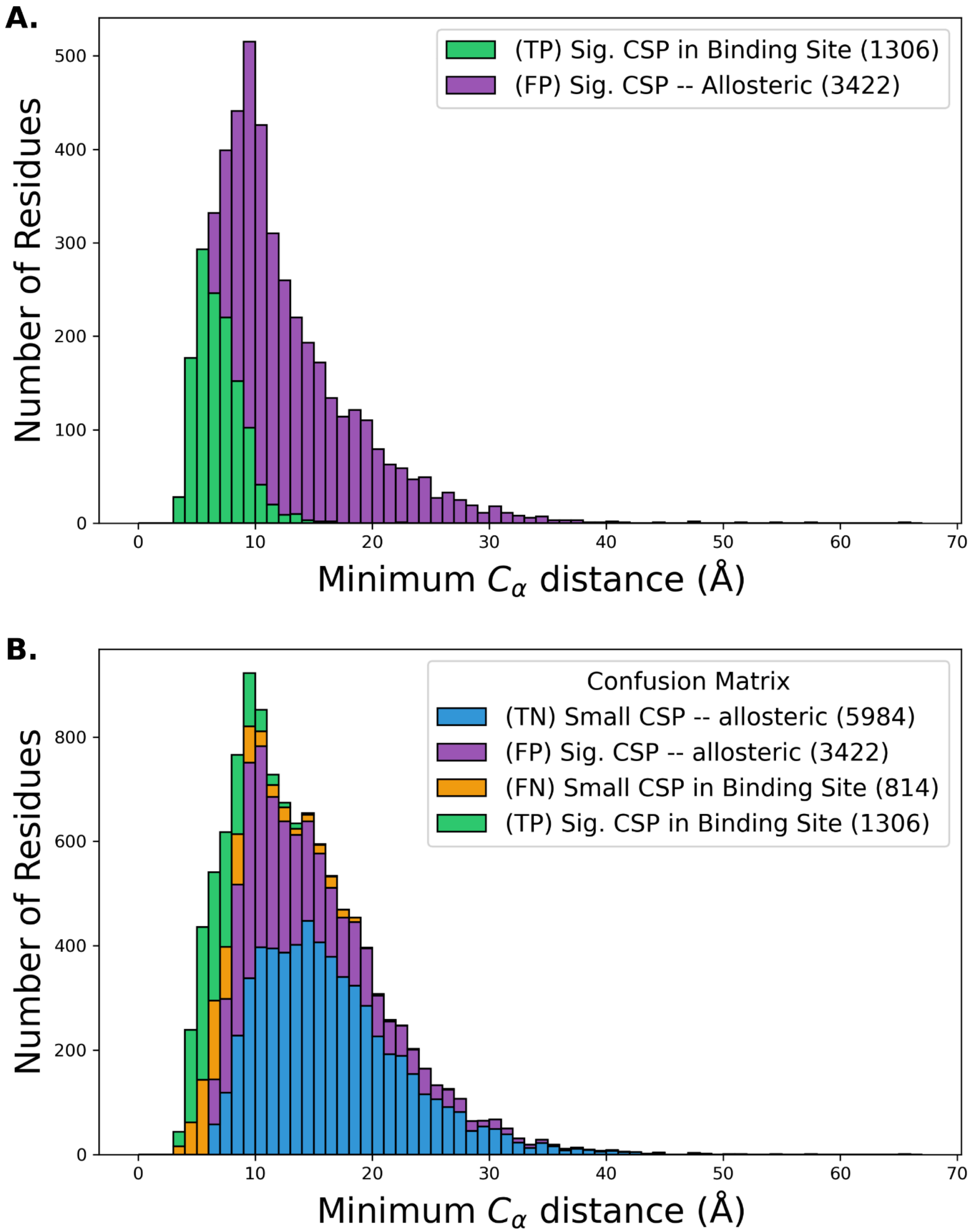
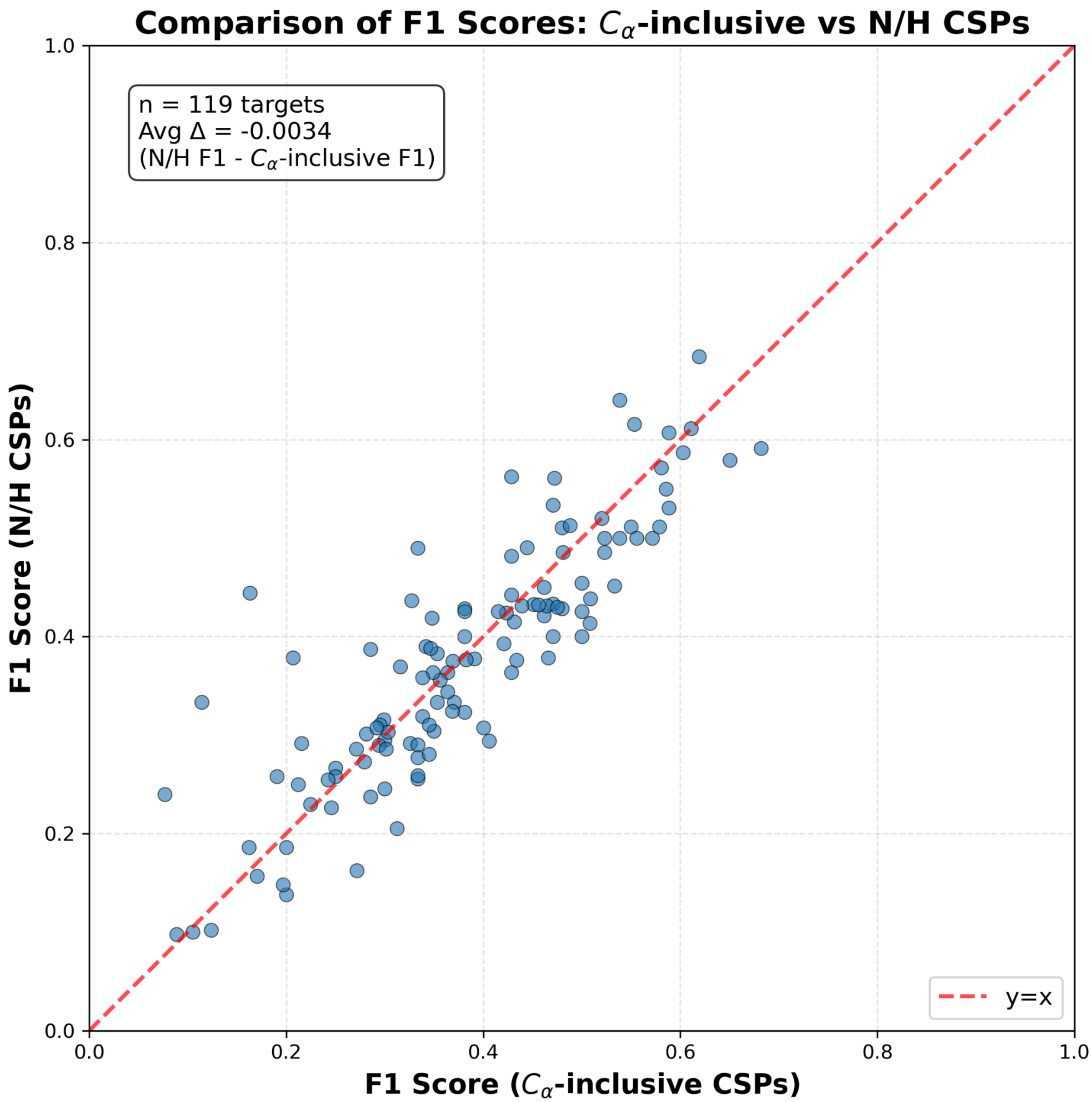
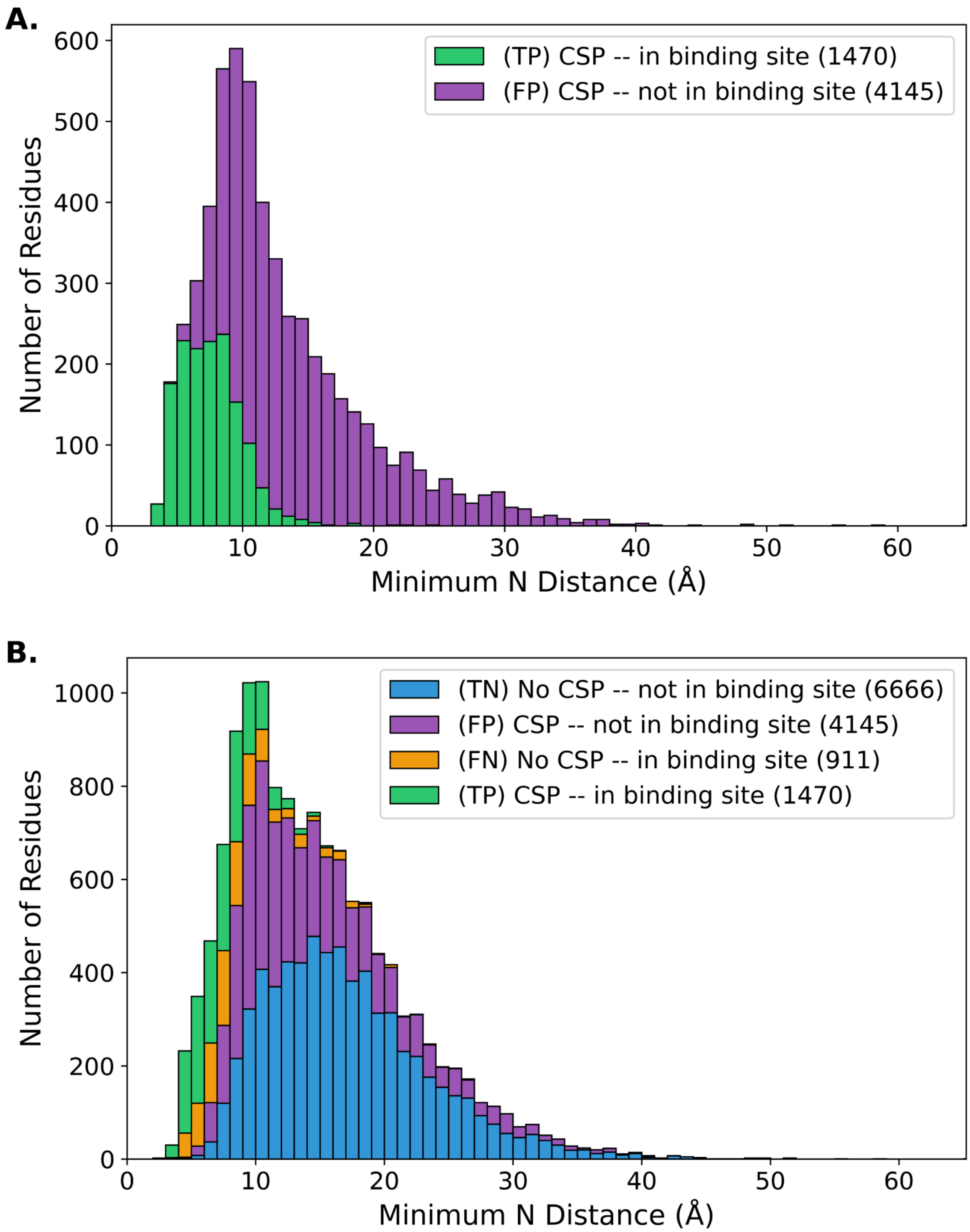


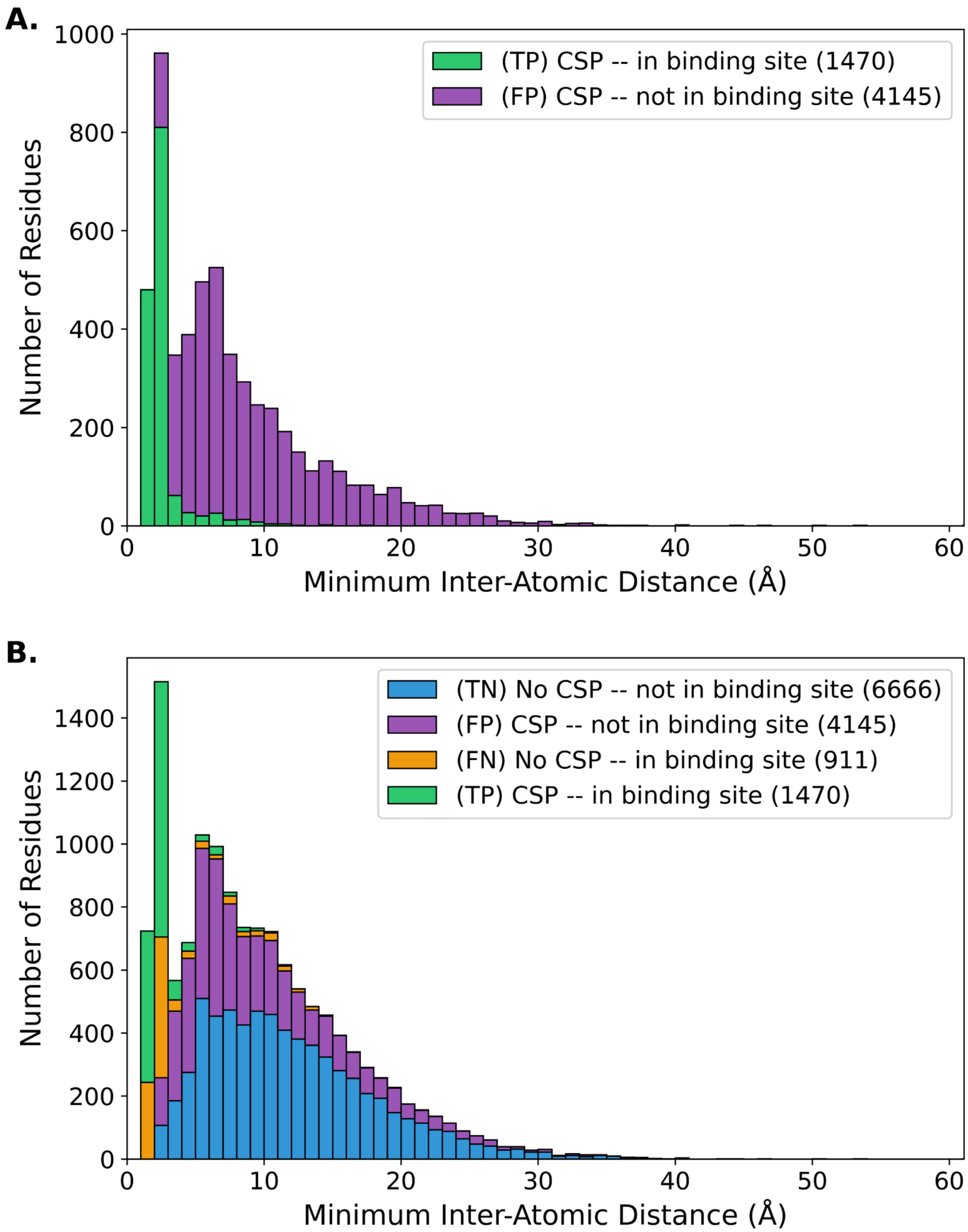

**SI Fig. S10**. **CA-inclusive CSP Confusion Matrix Histograms**. Calculating CSPs with CAs included in the formula does drastically not change the percentage of allosteric CSPs ~72%. CSPs are calculated using formula described in **Eqn. 2**.



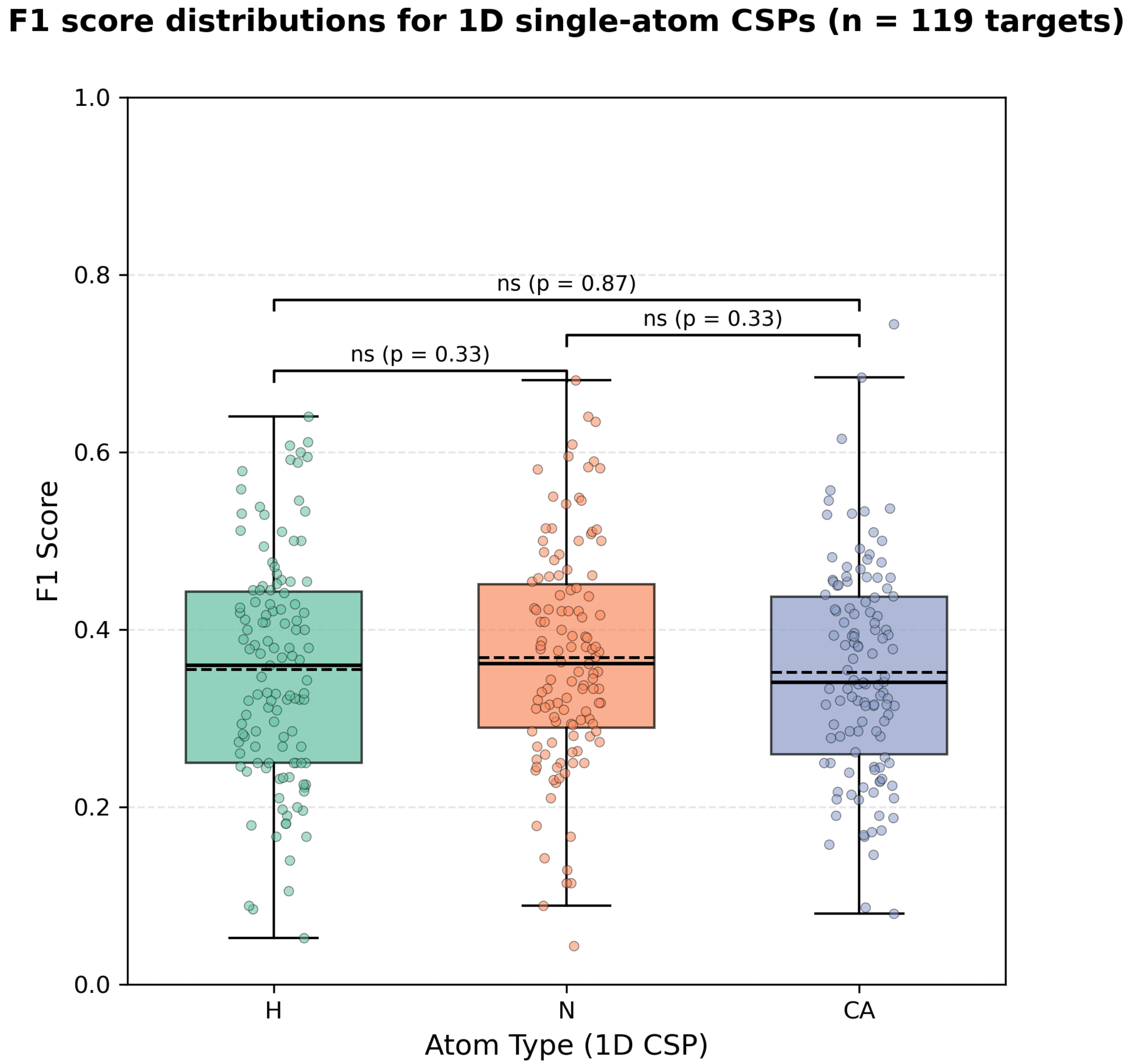

**SI Fig. S11. Change in F1 scores when considering CA-inclusive CSPs**. This plot shows that over the targets in the CSP DB, the average change in F1 score is negligible when comparing CSPs calculated from N/H shifts vs CSPs calculated with N/H/Ca shifts. CA-inclusive CSPs are calculated following **Eqn. 2**.



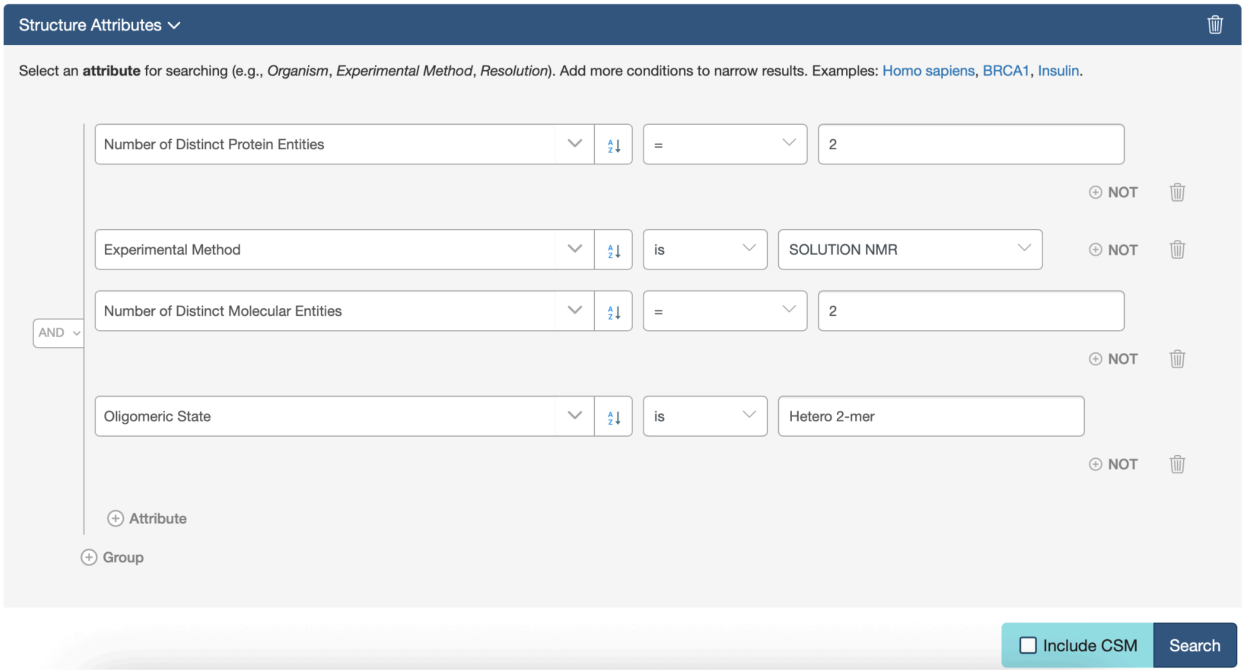

**SI Fig. S12. CSP DB Confusion Matrix Histograms by closest interchain N-N distance.**



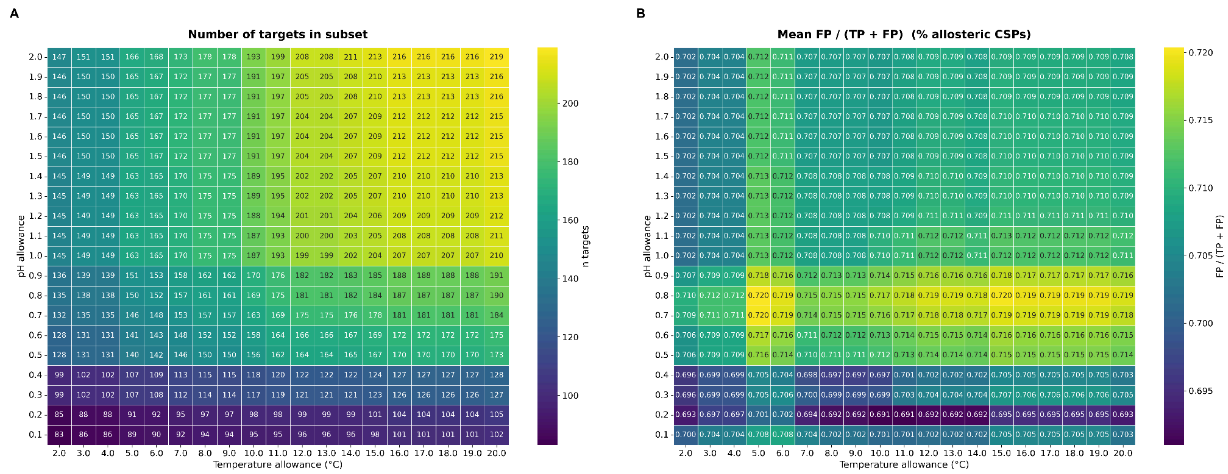

**SI Fig. S13. CSP DB Confusion Matrix Histograms by closest interchain atom-atom distance.**



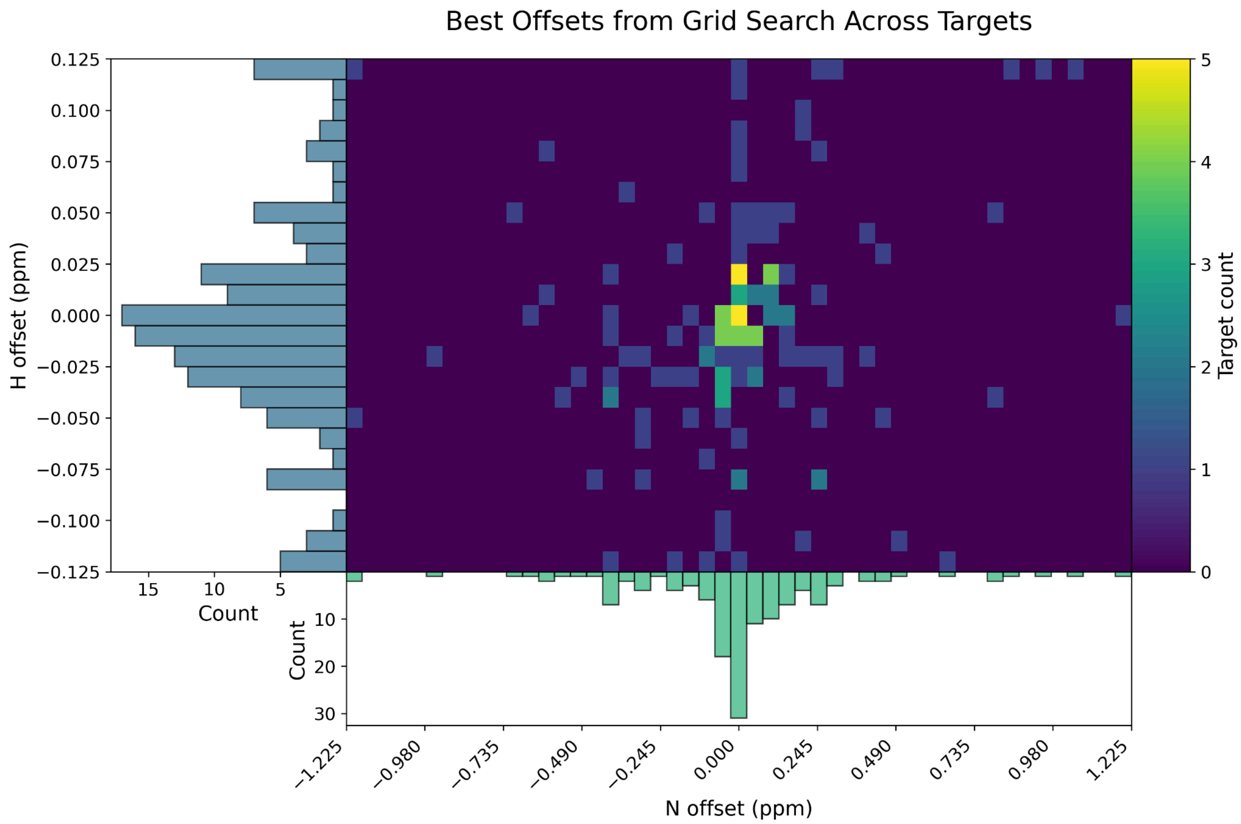

**SI Fig. S14. F1 Scores for 1D CSPs.** This plot shows there are no statistical differences between distributions of calculated F1 scores from 1D CSPs H/N/CA atom types. CSPs are calculated as root-squared differences after applying an ideal holo offset for a 3D H/N/CA offset grid (see **Methods** for details). Statistical analysis is performed using paired Wilcoxon tests with Holm correction.



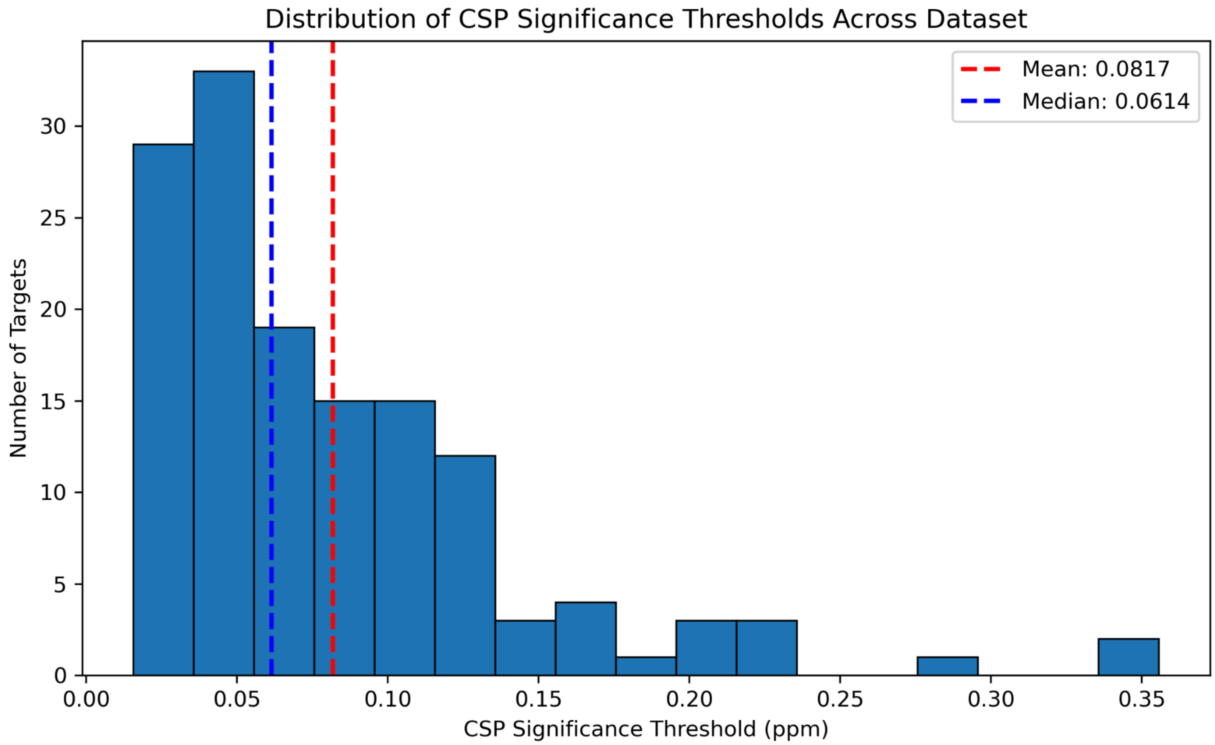

**SI Fig. S15. PDB Advanced Search Settings.** For the subset of targets manually scraped from the PDB, we restrict our analysis to these settings: entries resolved by Solution NMR, with two and only two distinct protein chains. At the time of curation, this advanced search resulted in 720 hits on the RCSB PDB. From this list of 720, entries were included in the CSP DB if there were N and H chemical shifts deposited in the BMRB for an identical sequence which was evaluated in both apo and holo experiments.



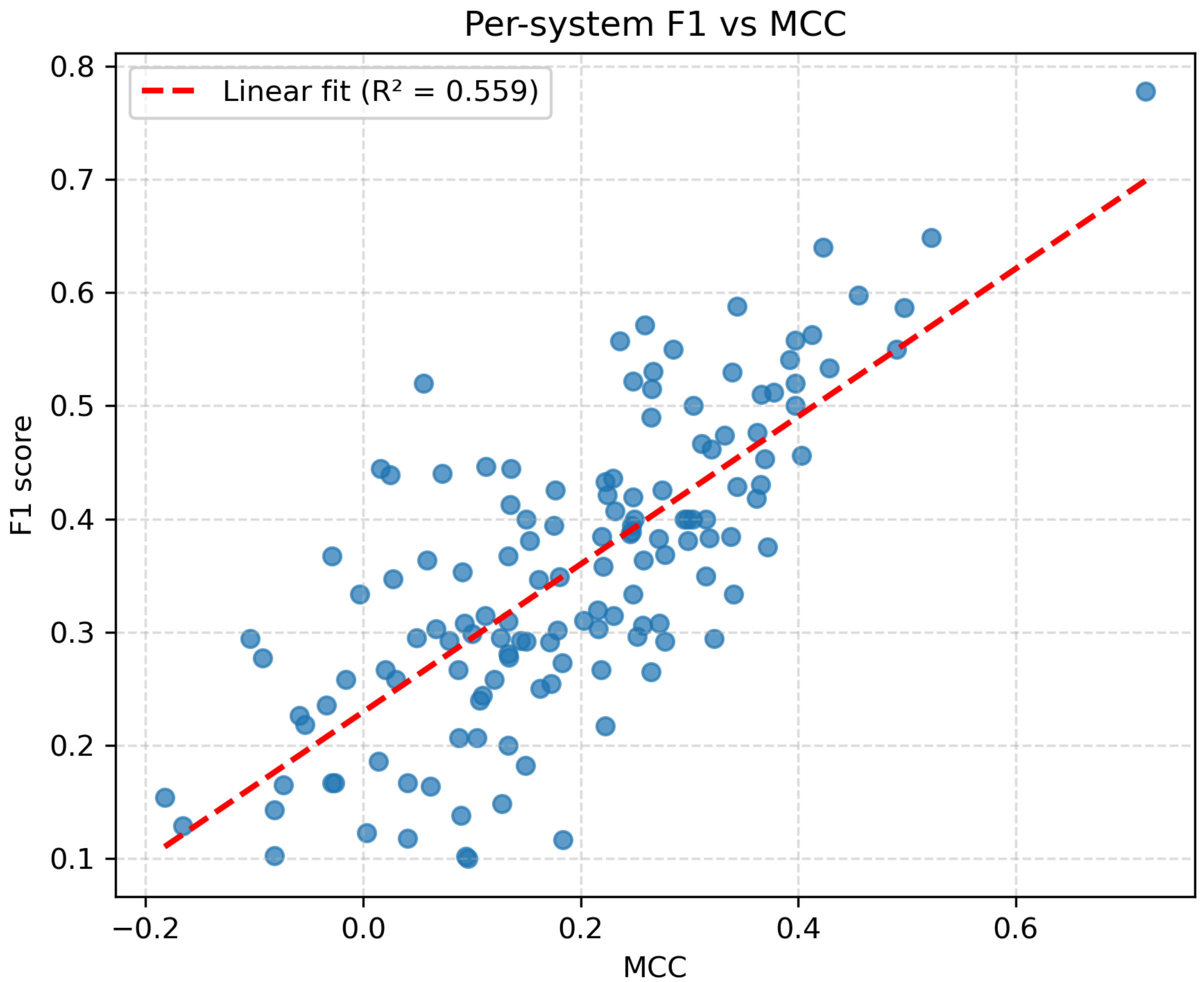

**SI Fig. S16. Impact of Experiment Condition Thresholding on Results.** These heatmaps demonstrate that changing the threshold of difference in pH and temperature between the apo and holo HSQC experiments does not have a strong effect on the reported outcomes of this study. 219 receptors have experimental condition differences within ± 2 pH units and ± 20 ºC, and the mean percentage of allosteric CSPs across this more lenient thresholding is only slightly less (70.6%) than the value reported in the main text on the subset with conditions ± pH 0.5 units and ± 5 ºC (71.6%).




**SI Fig. S17**. **Ideal offsets across the whole dataset for N/H**. This heatmap shows the offsets used to align the apo/holo HSQC plots. The majority of targets require small adjustments within 0.025 ppm on the Hydrogen axis and 0.245 ppm on the Nitrogen axis.




**SI Fig. S18. CSP significance threshold Histogram.** This histogram summarizes the thresholds for determining whether a CSP for a given receptor is significant. For a small number of targets, the calculated significance threshold is significantly higher than one would expect, we attribute this to differences in experimental conditions between the apo and holo NMR experiments which aberrantly perturbs all observed shifts. We prune from the CSPdb any targets with a calculated significance threshold > 0.6 ppm. This includes the following targets: 2LMC, 1OO9, 2L12, 2L00, 2N01, 2MP0, 8VOI, 8U2Y, 6OSW.




**SI Fig. S19. F1 vs MCC score comparison across CSP DB targets.** This plot shows the correlation between calculated F1 and MCC metrics. MCC is a more reliable metric for larger systems with many True Negatives (small CSPs outside of the binding site) (7).

**References**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